Using rsync to Back Up a System

Imagine you are trying to copy a large file to a remote server and the operation fails due to network connectivity. So you restart the process from scratch, but face the same issue again and again. rsync will come to your rescue here, as it uses the delta transfer algorithm to perform the copy operation effectively. In this article, we will see how to use rsync to create a backup of a system.

Rsync stands for 'remote sync'. This protocol has been designed and implemented by Andrew Tridgell, and can be used to synchronise files from local or remote systems. It uses the delta transfer algorithm to copy only the modified portion from the source to the destination file. This saves network bandwidth and effectively reduces the time taken to complete the operation.

Salient features and installation

a) *Efficiency:* rsync is efficient as it transfers only 'delta' instead of copying the whole file.
b) *Performance*: It performs better than scp because of the efficient delta transfer algorithm.
c) *Security:* It allows encryption of data using the SSH protocol during transfer.
d) *Ease of use:* It is simple to use. Also, no special privileges are required to use it.

To install rsync on RPM based GNU/Linux distros, execute the command given below with root privileges:

```
# yum install rsync
```

Next, execute the following command as a non-root user to verify its installation:

```
$ rsync -version
```

If you see an output similar to the following, then rsync has been installed successfully:

```
rsync  version 3.1.3  protocol version 31
    Copyright (C) 1996-2018 by Andrew Tridgell, Wayne
Davison, and others.
    Web site: http://rsync.samba.org/
```

Primitive operations

The syntax of rsync command is as follows:

```
rsync [options] <source> <destination>
```

In rsync, the source and destination can be any combination of local or remote systems. Please note that the syntax will vary when a remote system comes into the picture.

a) Sync a single file on the local server: Let us use rsync to synchronise a 1GB file on a local server:

```
[root]# ls -lrth
 total 1.0G
 -rw-r--r--. 1 root root 1.0G Dec  6 17:46 data.iso

  [root]# rsync -vh  --no-W data.iso /opt/backup/
```